Module 1
NFS Fundamentals

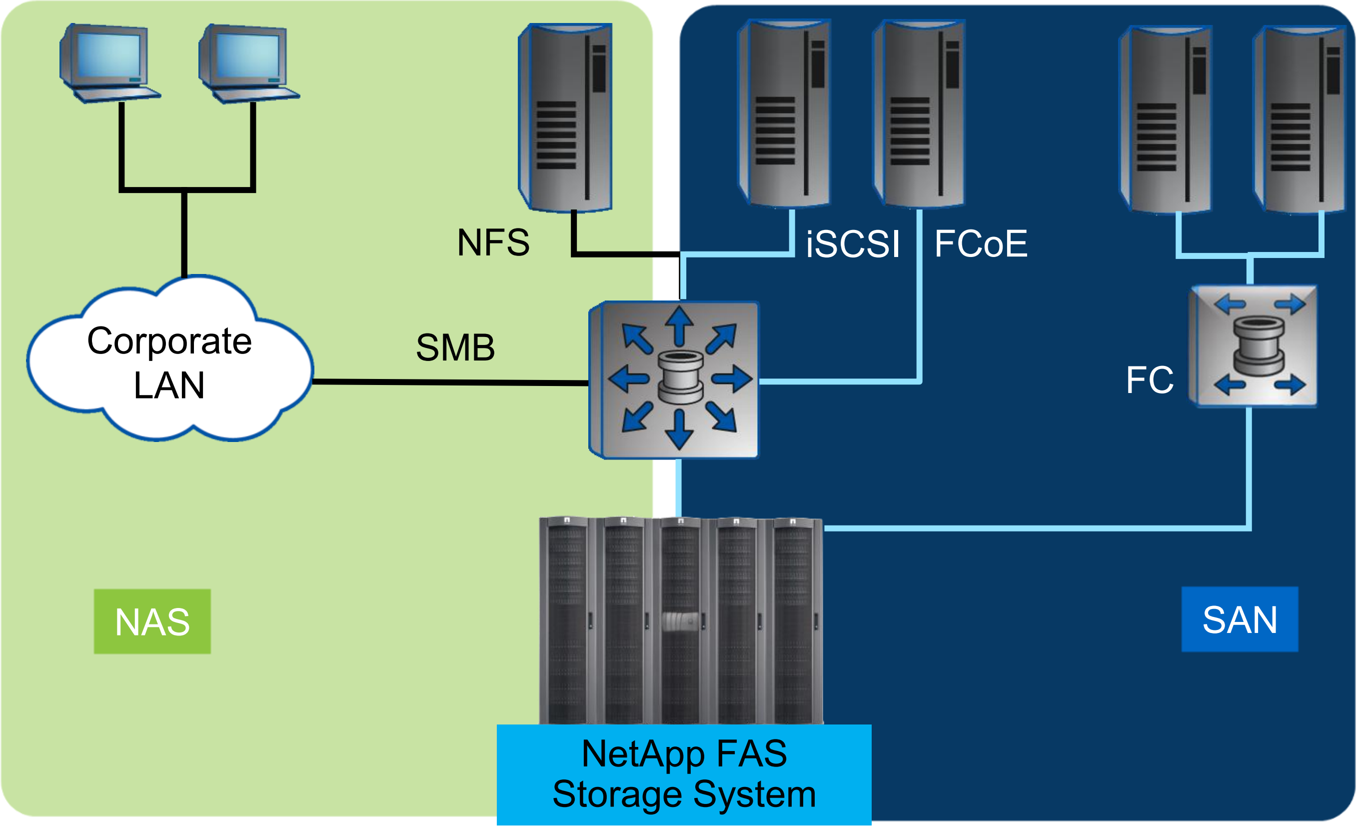
About This Module

This module focuses on enabling you to do the following:

- Explain the differences between the NFS protocol and SAN protocols
- List NAS namespace architectures and namespaces
- Summarize client access to NFS data in ONTAP software
- List the NFS enhancements for different releases of ONTAP software

Lesson 1
Overview of ONTAP NAS File Access

Comparing NAS and SAN



What are NAS and SAN?

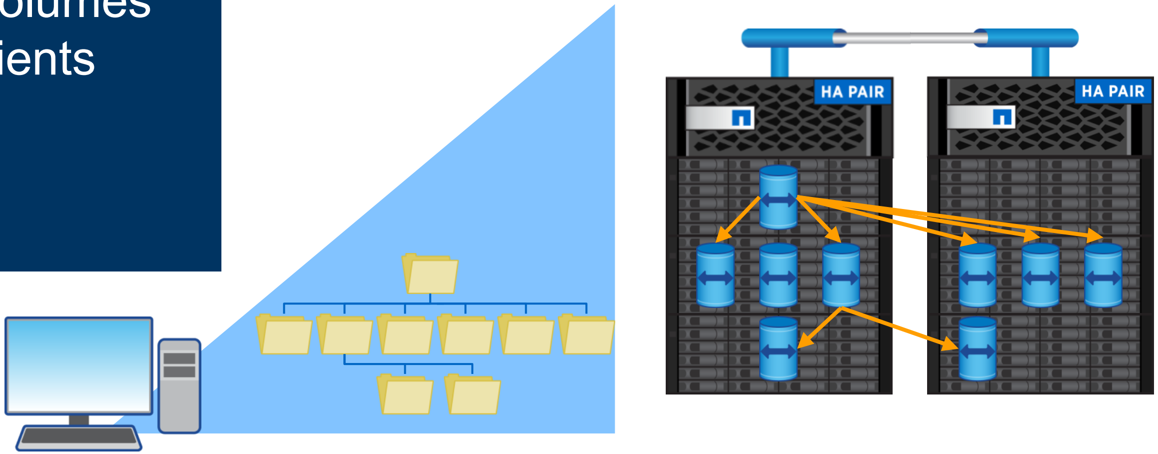
What is the difference?

Namespace Architecture: Namespaces and Junction Points

- A namespace for NAS is a logical grouping of volumes that are connected to create a hierarchal file system.
- Junction points are used to connect volumes into the namespace.

Junctions:

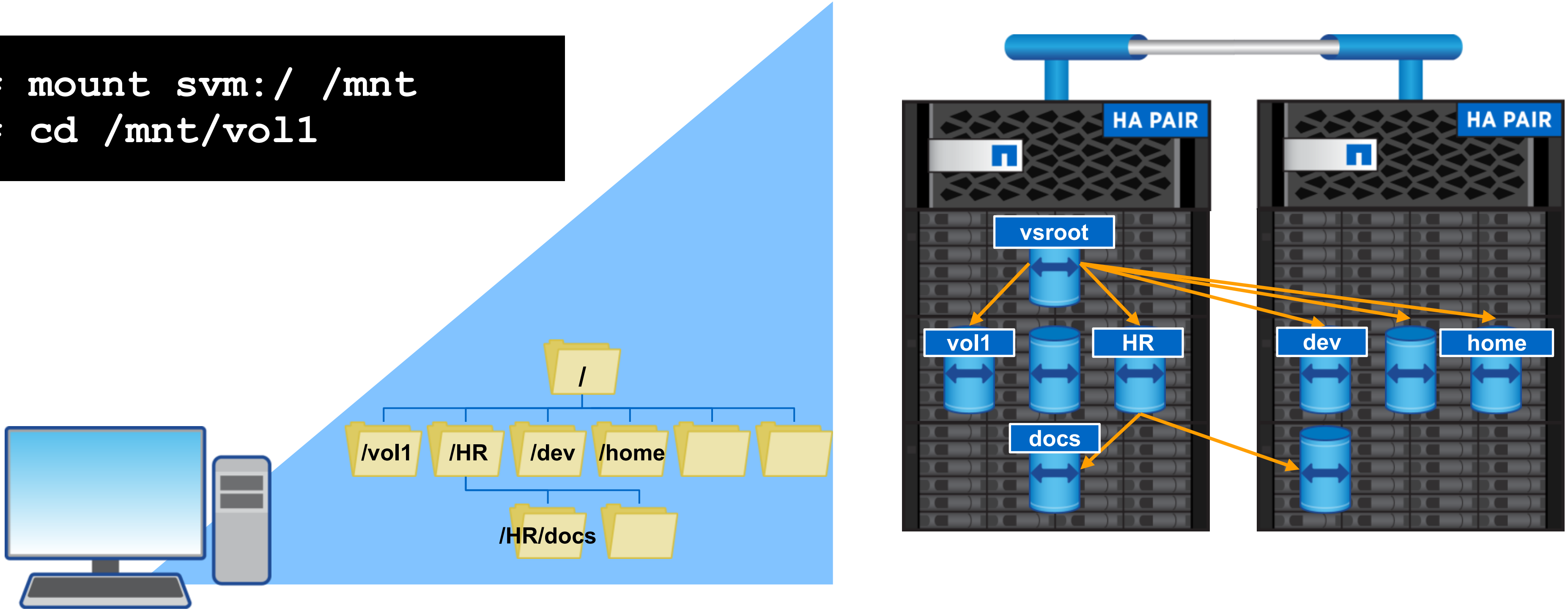
- Are volumes mounted to volumes
- Appear as directories to clients
- Can span entire clusters
- Are managed by storage administrators



Accessing the Namespace

The storage virtual machine (SVM) root is the entry point into the namespace.

```
# mount svm:/ /mnt
# cd /mnt/vol1
```



Namespace Architecture: Single Branched Tree

Access paths

- The SVM root is the entry point into the namespace.
- The junction point can be a volume or directory beneath the root.
- All other volumes are mounted beneath the top junction point.
- This example shows a single branched tree where the entry point to the namespace is a single directory that is named “data.”

```
Vserver      Volume      Junction Active    Junction Path            Junction Path Source
---------    -------     ---------------    --------------           --------------------
vs1          customer1    true              /data/customer1          RW_volume
vs1          customer2    true              /data/customer2          RW_volume
vs1          project1     true              /data/project1           RW_volume
vs1          project2     true              /data/project2           RW_volume
vs1          vs1_root     true              /                        -
```

Namespace Architecture: Multiple Branched Trees

Access paths

Vserver	Volume	Junction Active	Junction Path	Junction Path Source
vs1	audit	true	/audit	RW_volume
vs1	audit_logs1	true	/audit/logs1	RW_volume
vs1	audit_logs2	true	/audit/logs2	RW_volume
vs1	audit_logs3	true	/audit/logs3	RW_volume
vs1	sales	true	/data/eng	RW_volume
vs1	mktg1	true	/data/mktg1	RW_volume
vs1	support	true	/data/support	RW_volume
vs1	project1	true	/projects/project1	RW_volume
vs1	project2	true	/projects/project2	RW_volume
vs1	vs1_root	true	/	-

Namespace Architecture: Multiple Standalone Volumes

Access paths

- Multiple access points are junctioned into the SVM root volume or a directory within the root volume.
- Each junction point is a volume.
- Each volume is a unique path.

Vserver	Volume	Junction Active	Junction Path	Junction Path Source
vs1	eng	true	/eng	RW_volume
vs1	mktg	true	/dept/mktg	RW_volume
vs1	project1	true	/project1	RW_volume
vs1	project2	true	/project2	RW_volume
vs1	sales	true	/sales	RW_volume
vs1	vs1_root	true	/	-

Authentication Versus Authorization

Who you are versus what you can do

Authentication	Authorization

▪ To gain access to a NAS share, users must prove who they are.
▪ This information helps ONTAP to determine what access the user has.
▪ Accounts can be local to the cluster or can be remote accounts that are stored in name servers:
    ▪ Remote users are stored on Active Directory, Lightweight Directory Access Protocol (LDAP), or Network Information System (NIS).
    ▪ Name services are used to locate host names, users, groups, and netgroups.
▪ Authentication can include Kerberos, for more secure NFS connections.
	▪ After a user has authorization and has an identity, that identity is then compared against access control lists (ACLs) to determine what permissions the user has.
	▪ The determination of whether the entity is authorized to perform the requested action is based on permissions that are configured using the following:
	    ▪ Export policies
	    ▪ Storage-Level Access Guard
	    ▪ File-level security on the object set via NTFS or Unix from the client

Export Permissions versus File Permissions

Different access rights for different access points

Export Permissions

- Export permissions control who can access NFS mounts.
- Export permissions are set for hostname/IP/subnet/netgroups:
  - Export permissions don’t act at the user/group level.
  - Subnet can be specified using the Classless Inter-Domain Routing (CIDR) address.
- Exports are grouped into policies.
- Access is controlled by export policy rules.
- Multiple clients can live in the same rule (comma-separated or via netgroup).
- Policies can have many rules.
- Rules control the following:
  - Read/write access
  - Root access
  - Anonymous access
  - Protocol access (NFS version)
  - Security flavor access (sys, krb5*)

File Permissions

- File permissions control who can access files or folders.
- File permissions are determined by who the user is authenticated as, and what security style is in use.
- File permissions follow standard rules:
  - File permissions always win.
  - Even if a client has been granted write access, file permissions can override.
  - Explicit denies always override reads/writes.
  - UNIX can use mode bits (777) or NFSv4.x ACLs.
    (ACL support must be enabled on the storage system.)
- NFS permissions are set from the clients.

Export Policy Caches and Checking

- `export-policy access-cache`

  The access cache stores entries that consist of one or more export rules that apply to clients that attempt to access volumes or qtrees.

- `export-policy check-access`

  Enables you to test export policies to ensure that they work as intended and to troubleshoot client access issues.

- `export-policy config-checker`

  Enables you to run a background job that checks export policy configuration and reports errors for each affected rule.

Credential Cache

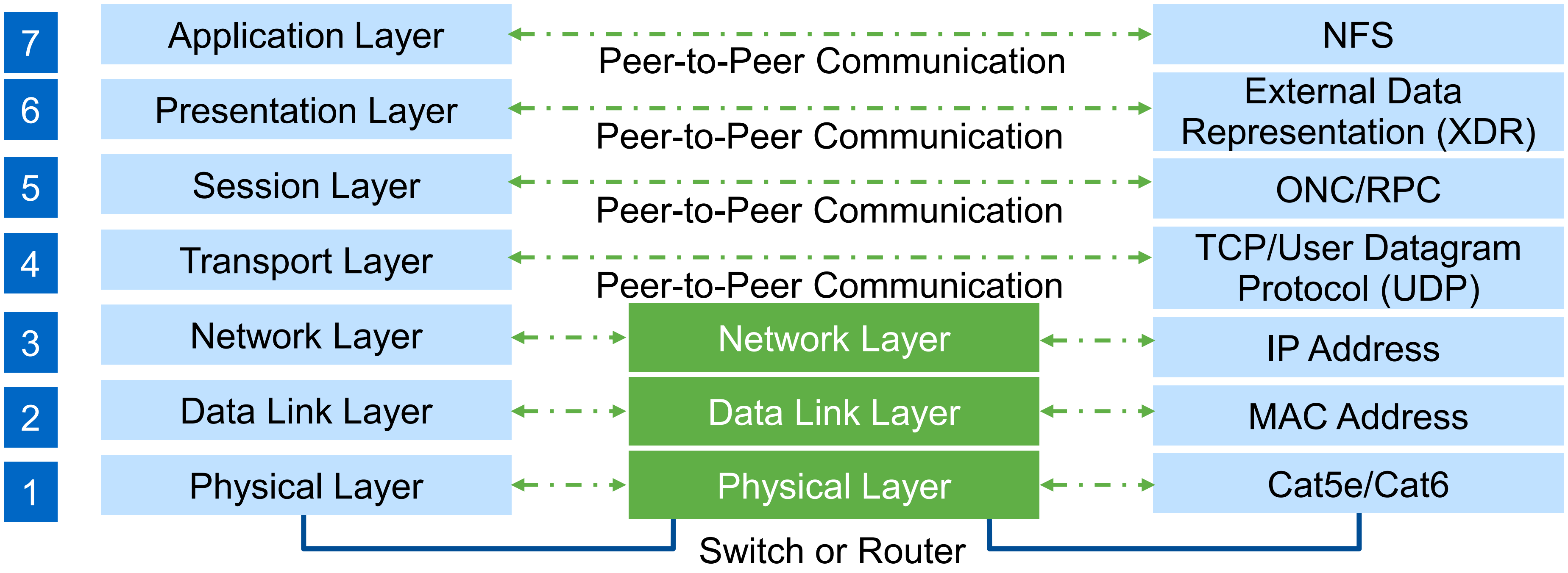
- With the credential cache:
  - ONTAP software retrieves user credentials to authenticate the user.
  - Successfully processed credentials are stored on the node for 24 hours after authentication, for later reference.
  - Denied access credentials are stored for two hours.
  - The credential cache eases the load by minimizing requests to and from external name services.
  - The credential cache speeds up authentication processes, because credentials do not have to be retrieved from external name services.
- Without the credential cache, ONTAP software queries name services every time file access is requested.

Lesson 2
Summarizing NFS Client Access on ONTAP

NFS

- NFS enables networked computers to access shared files.
- Almost all major operating systems support NFS:
  - Linux
  - Solaris
  - HPE UX
  - AIX
  - Windows
- Transport occurs over IPv4 or IPv6.

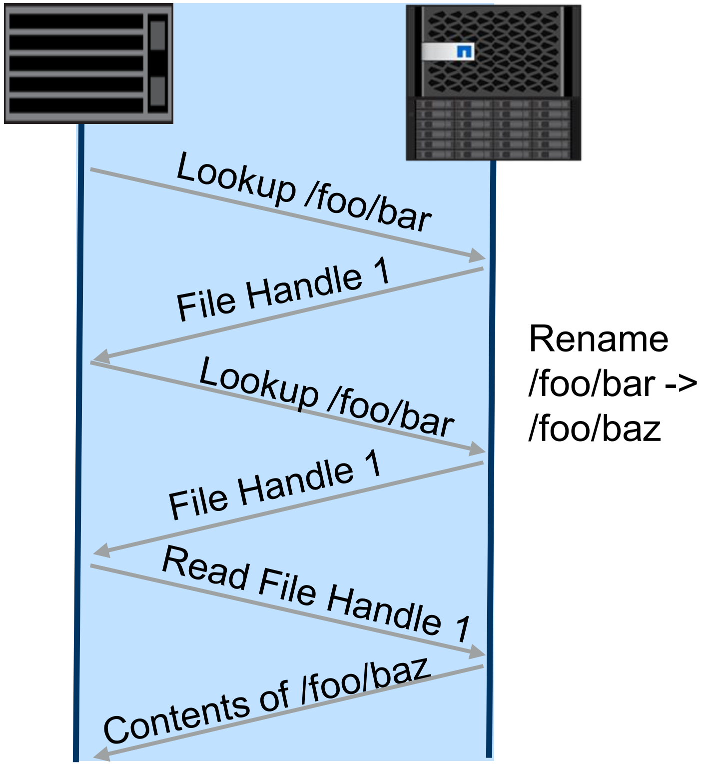
NFS and the OSI Model



ONC/RPC = Open Network Computing/Remote Procedure Call | OSI = Open Systems Interconnection | NFSv3 = NFS version 3

NFSv3 Client-Server Communication

- File handle:
  - A basic unit of NFS file access
  - A client reference for a file on the server
  - Created on the server
  - Used by the client
- Properties:
  - Isolation between client and server:
    - Preserves access through client disconnect and reconnect
    - Increases availability
  - Path name independence:
    - Survives renames
    - Increases flexibility

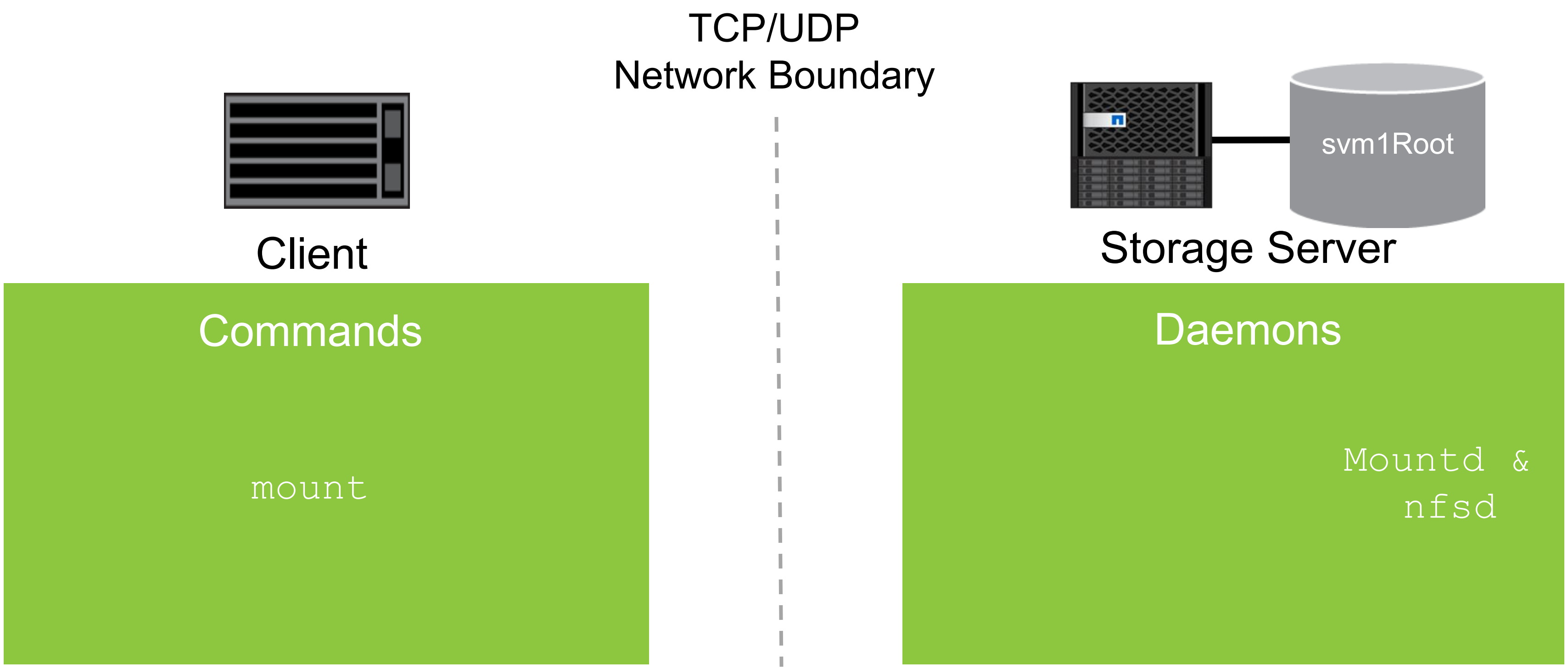


NFSv3 Example

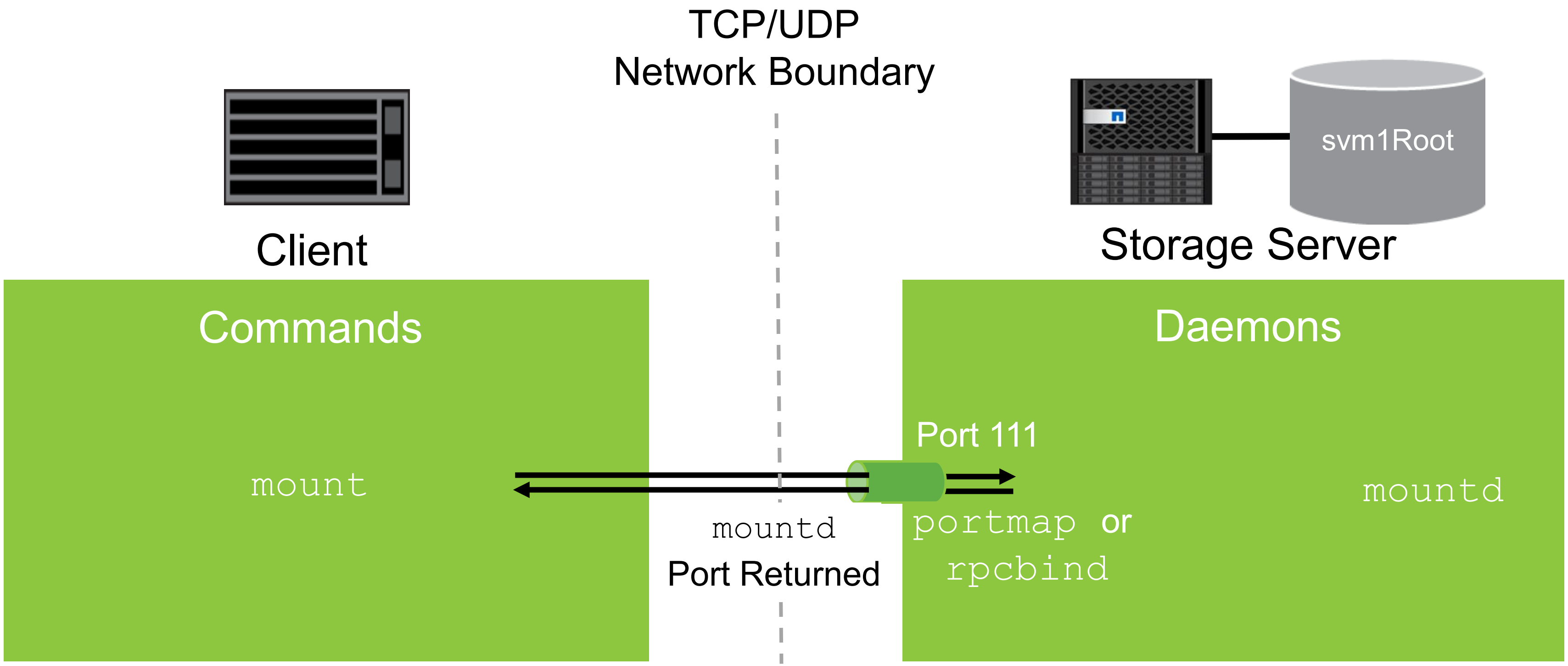
NFSv3 Ancillary Protocols

- Network Lock Manager (NLM):
  - Advisory locks
  - Synchronous and asynchronous locking
  - Locks that the client explicitly releases
  - Lock state that is maintained at the client and server by using `lockd`
- Network Status Monitor (NSM):
  - Facilitates NLM lock recovery
  - Enables the client and server to maintain a state number by using `statd`
  - Provides notification of loss of server and client state

NFSv3 Client and Server

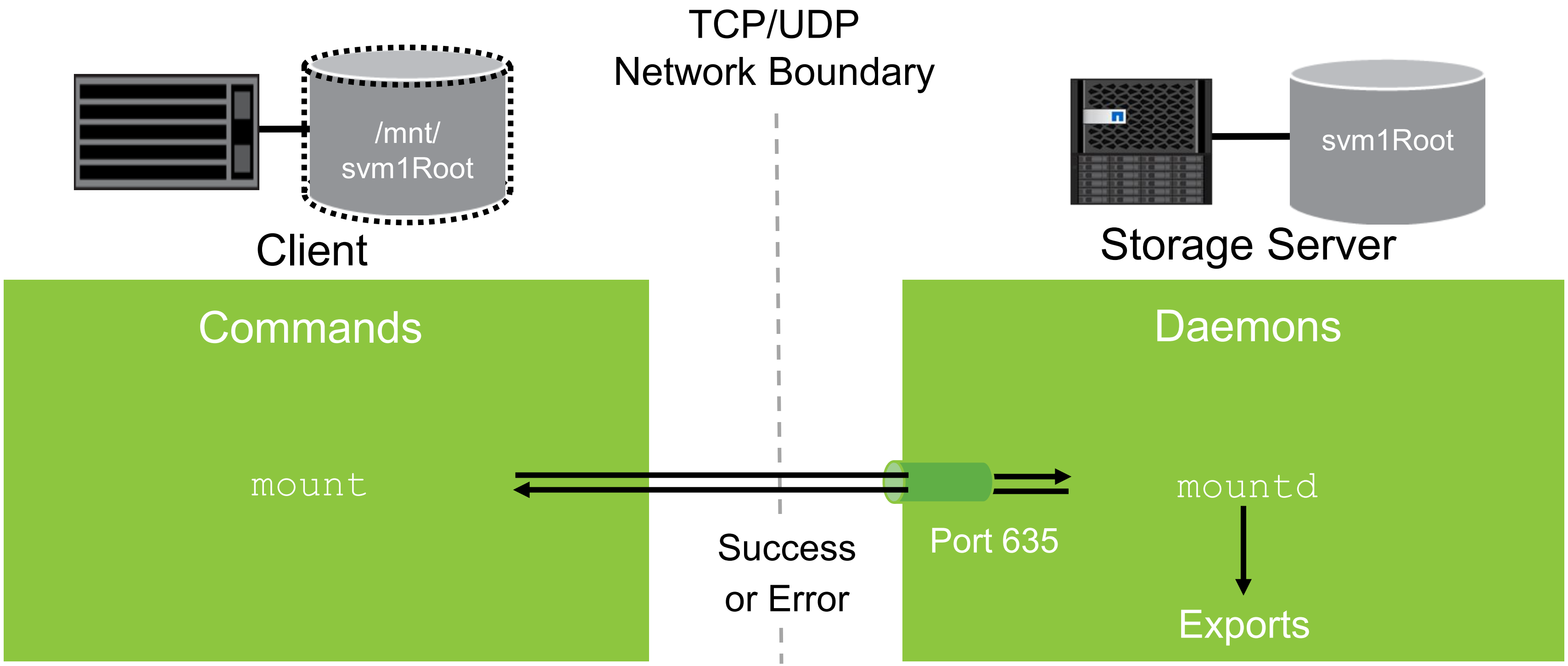


NFSv3 Mount Daemon Port Lookup




```
# mount svm1:/ /mnt/svm1Root
```

NFSv3 Mount Request

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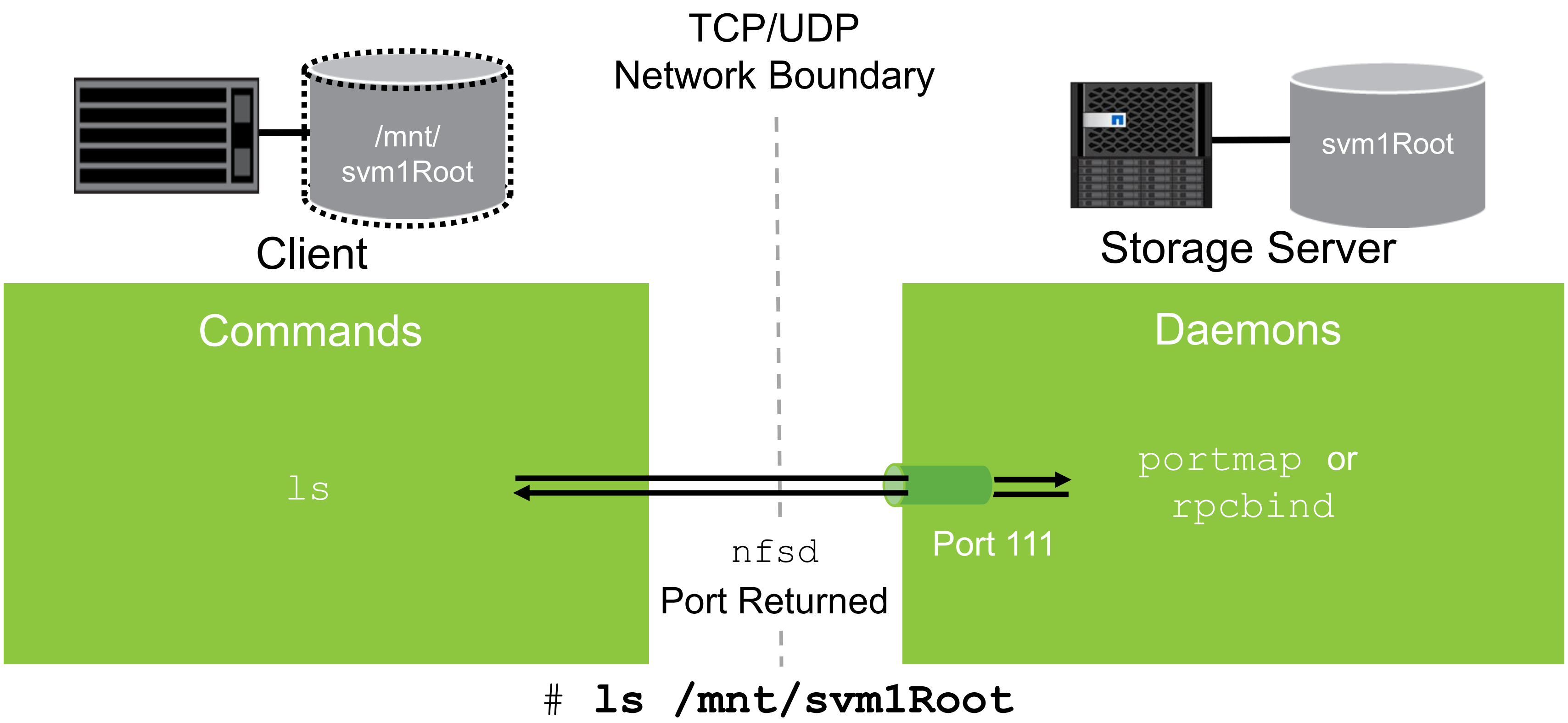
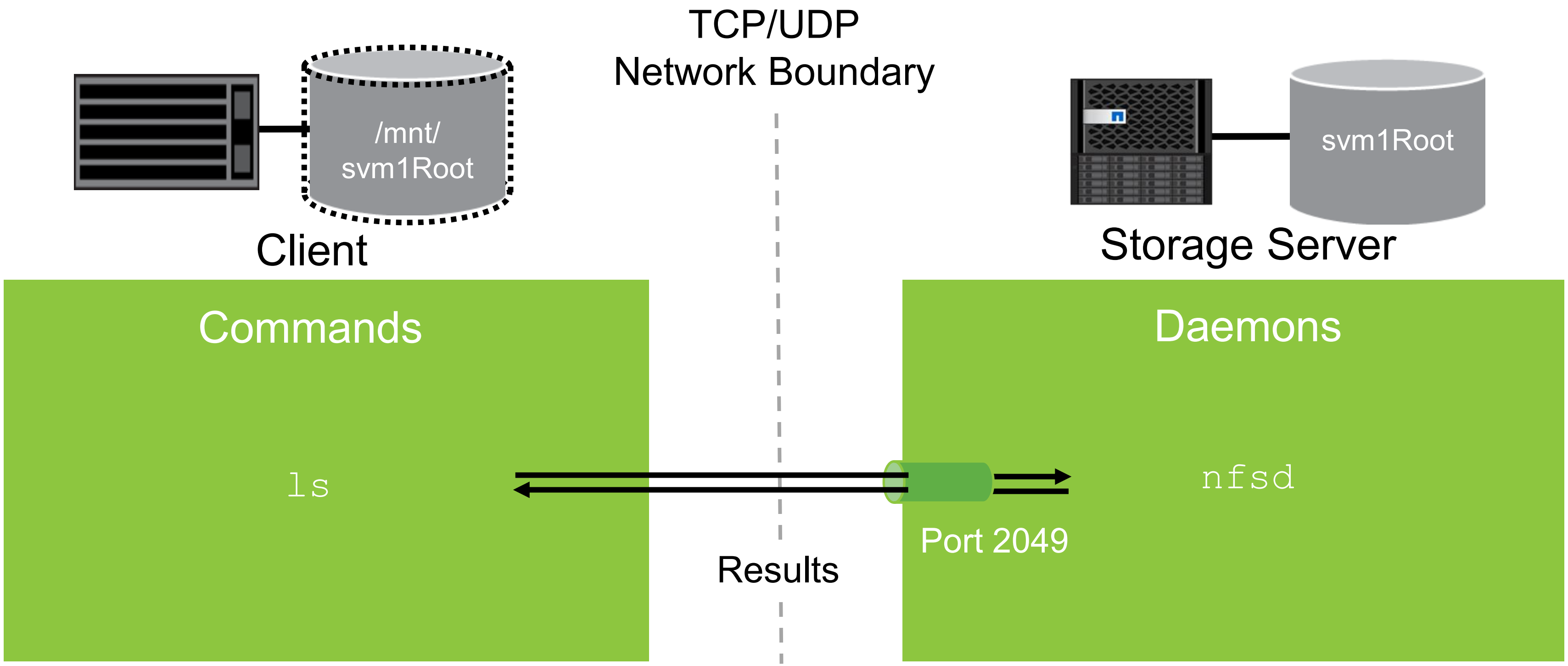
NFSv3 NFS Call Daemon Port Lookup



NFSv3 NFS Call Command Request




```
# ls /mnt/svm1Root
```

Lesson 3
NFS Protocol Versions and Enhancements

ONTAP NFS Support Matrix

	Clustered Data ONTAP 8.1.x	Clustered Data ONTAP 8.3.x	ONTAP 9
NFSv3	Yes	Yes	Yes
NFSv4	Yes	Yes	Yes
NFSv4 with Delegations	Yes	Yes	Yes
NFSv4.0 with Referrals	Yes	Yes	Yes
NFSv4.1	Yes	Yes	Yes
NFSv4.1 pNFS	Yes	Yes	Yes
NFSv4.1 with Referrals	Yes	Yes	Yes
NFSv4.1 with Delegations	No	Yes	Yes
NFSv4.1 parallel NFS (pNFS) with Delegations	No	No	No

NFS Features in ONTAP Software

- The ~snapshot directory can be hidden from NFSv3 clients.
- A config-checker job can be run that records rule violations in an error rule list.
- NFS clients can view a list of exports that are available by using the `showmount -e` command.
- Multiple client match specifications are present in an export rule.
- Access cache parameters for NFS exports can be specified for individual SVMs.
- An access cache that is not actively in use is no longer refreshed.
- SnapMirror Synchronous supports NFSv4.0 and NFSv4.1 and mixed protocol support for NFSv3 and SMB.
- Auditing support with NFS security tracing adds permission tracing filters, to log why NFS servers allow or deny a client request.

NFS Features

Kerberos and export policies

- Enhancements for Kerberos 5:
  - Authentication with privacy service (krb5p) is supported.
  - Privacy service verifies the integrity of received data, authenticates users, and encrypts data before transmission.
- Multiple client match specifications:
  - Specify multiple client match values in every rule and in the NFS export policy.
  - Provide functionality that is comparable to the standard `/etc/exports` file in Data ONTAP operating in 7-Mode.
- NFS export access cache enhancements:
  - Access cache parameters for NFS export rules can be specified for individual storage virtual machines (SVMs).
  - Inactive access cache entries are no longer refreshed, which improves efficient communication.

NFS Features

Kerberos and export policies

- Differentiated Services Code Point (DSCP) marking is supported:
  - Is also known as quality of service (QoS) marking
  - Is a mechanism for classifying and managing network traffic
  - Supports NFS
  - Is enabled by providing an IPspace, protocol, and DSCP value
  - Is a component of Unified Capability (UC) compliance
- IPv6 support for Dynamic DNS is available on IPv6 LIFs.

Configure and enable DSCP

- **View Current Configurations:**
  > network qos-marking show

- **Enable or Disable Marking:**
  > network qos-marking modify -ipspace Default -protocol NFS -is-enabled true

- **Change the DSCP Value:**
  To set a specific DSCP number (e.g., 20) for a protocol:
  > network qos-marking modify -ipspace Default -protocol NFS -dscp 20

NFSv4 and Beyond

- Ancillary protocols are not needed: Portmapper, mount, NLM, and NSM are fused into one protocol.
- A firewall-friendly approach uses well-known port 2049 (no portmapper).
- Stronger security is recommended.
- Caching semantics (delegations) are improved.
- NFSv4.1 and later include added sessions and parallel NFS (pNFS).

Module Summary

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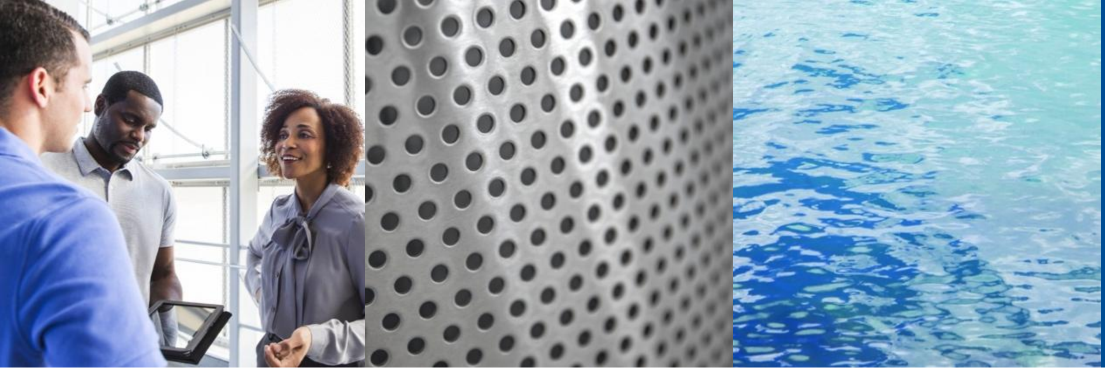

Take a Poll

Two-Question Quiz

Duration: 15-20 minutes

Poll Questions

1. Which statement about export policies and rules is false?
   a. Export policies control who can access NFS mounts.
   b. Export policies control who can access individual files and folders.
   c. Export rules are grouped into policies.
   d. Export policies can have more than one rule.

Poll Questions: Answers

1. Which statement about export policies and rules is false?
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   d. Export policies can have more than one rule.

Poll Questions

2. How long are successfully processed credentials held by default in the credential cache?
    a. 2 hours
    b. 12 hours
    c. 24 hours
    d. indefinitely

Poll Questions: Answers

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Poll Questions

1. NFS operates at what layer of the OSI model?
   a. Presentation
   b. Session
   c. Application
   d. Data Link

Poll Questions: Answers

1. NFS operates at what layer of the OSI model?
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Poll Questions

2. Which basic unit of NFS is used for file access?
    a. lock
    b. NT LAN Manager (NTLM)
    c. `ready` command
    d. file handle

Poll Questions: Answers

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Poll Questions

1. What version of NFS introduces support for parallel NFS (pNFS)?
   a. NFSv4
   b. NFSv3
   c. NFSv4.1
   d. NFSv2

Poll Questions: Answers

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Poll Questions

2. What version of NFS is stateless?
    a. NFSv3
    b. NFSv4
    c. NFSv4.1
    d. parallel NFS (pNFS)

Poll Questions: Answers

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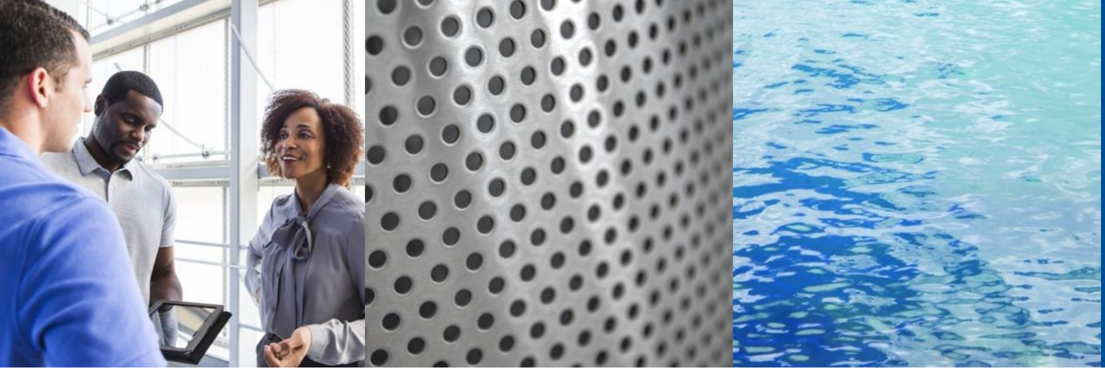

Complete an Exercise

Module 1: Checking the Exercise Environment

Duration: 15 minutes